



Yale University Department of Music

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Author(s): Paul Pedersen

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The Mel Scale

PAUL PEDERSEN

This paper investigates one finding in the field of experimental psychology and suggests the possibility of its use in musical composition.† In general, the perceived magnitude of physical variables does not correspond directly to a physical measure. Psychological scales have been constructed on the basis of experimental procedures to reflect the actual relationship between the perceived magnitude and the physical measure in quantitative terms. The mel scale is one such effort, where psychological pitch is related to frequency so that equal steps in mels (the unit of psychological pitch) are equal in subjective

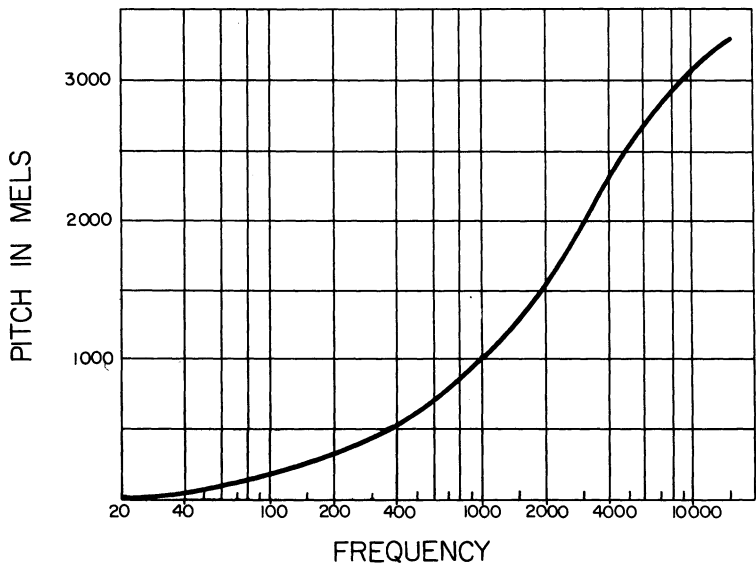
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size, and perhaps equal in perceptibility, throughout the audible frequency range.

Stevens, Newman and Volkman*1 and later Stevens and Volkman*2 presented the mel scale in 1937 and 1940. The experimental procedure must be considered to see how numbers are applied to purely subjective events like psychological pitch in the construction of a psychological scale such as the mel scale. Two different experimental methods were used to define the mel scale in the revised version presented in the 1940 paper. In the first technique a tone of fixed frequency was presented to listeners and they were required to set a variable tone to a pitch half as high. An arbitrary number was assigned to the first fixed frequency and then another number half as large was assigned to the average frequency of the pitches judged half as high. In practice, for instance, the first standard frequency, a sine-tone of 1000 cycles per second, was assigned the number 1000. Then the frequency which listeners judged on the average to be half as high was assigned the number 500. That is to say, the listeners were required to divide the interval between the standard frequency and a hypothetical zero pitch into subjectively equal parts. In spite of the fact that the concept of zero pitch is a rather difficult one, the subjects were able to perform the task consistently. The actual experiments used many different standard frequencies but the procedure and reasoning were as outlined here. The results of these experiments were corroborated by a second method which eliminated the problem of the zero pitch concept. In this procedure the listeners were given two fixed frequencies; for example, 200 and 6500 cps. They were required to set three variable tones to pitches which appeared to divide the interval between the two fixed tones into four subjectively equal parts. Three overlapping ranges, covering most of the audible frequency range, were used for the standard intervals. The results of this experimental procedure correlated well with the results of the first procedure. A scale was then formed by assigning equal subjective pitch intervals to equal distances on the ordinate of a graph where pitch in mels (ordinate) was plotted against log-frequency (abscissa). As a fixed point, 1000 mels was set equal to 1000 cps. Zero mels was set, by extrapolation, at 20 cps. The resulting graph (Figure 1) presents a curve on which a given number of mels corresponds to each physical frequency. It should be noted that in these experiments, the tones were heard successively and not simultaneously. Thus this scale refers to melodic and not harmonic pitch distances. A similar scale for harmonic intervals has been constructed by Feldtekeller and Zwicker*3.

figure

1



Thus the mel scale is, in the first place, a scale of subjectively equal pitch distances measured in equal numbers of mels. For example, the subjective distance between pitches of 200 and 300 mels (i. e. a 100 mel interval) should be subjectively equal to the distance between pitches of 2600 and 2700 mels (also a 100 mel interval). That is, equal size intervals measured in mel scale reflects the fineness of discrimination between frequencies. For example, a 10 mel step should be equally perceptible throughout the audible frequency range. Discrimination is measured by means of the difference limen, or DL, which is defined as the smallest frequency change that can be discriminated. Stevens and Davis conclude that "all DL's for frequency are of equal subjective magnitude." *4 This conclusion comes from separate frequency discrimination experiments, the results of which are consistent with the results of the mel scale. Thus DL's for frequency are of a constant mel size for a given loudness level, and a given size of mel interval will contain a constant number of DL's. Or in other words, equal mel steps are equally perceptible throughout the audible frequency range. Since it might be of some interest to know the approximate size of the DL for frequency at a moderate loudness level, the author has calculated, from the data given by Stevens and Davis *5, that the DL for a loudness level of 60 db is between 2 and 3 mels. According to Stevens and Davis *5 the size of the DL is constant above 60 db but increases in size as loudness decreases. If the loudness level drops to a very low level such as 5 db, the size of the DL increases to approximately 7 to 8 mels.

Table 1 was empirically obtained by a computer analysis of Steven's published curve of the mel scale (1940). A set of coordinates, mels vs. frequency, was taken from the published graph and fed into an IBM 7094 computer programmed to generate an equation which best fitted these coordinates. The resulting equation, which was then used to produce the table, is as follows:

$$\begin{aligned} \text{FREQUENCY} = & 0.146750532 \times 10^{-12} (\text{MEL})^5 - 0.795481794 \times 10^{-9} (\text{MEL})^4 \\ & + 0.152864002 \times 10^{-5} (\text{MEL})^3 - 0.687099785 \times 10^{-3} (\text{MEL})^2 \\ & + 0.805045046 (\text{MEL}) + 2.14597315 \end{aligned}$$

A word of explanation concerning the format of the table may be helpful for those unacquainted with a conventional log-table format. The left hand column is mels in ten-mel steps from 0 to 3100 mels. The other ten columns are the frequencies corresponding to mels taken in one-mel steps. For example, here

is the row beginning with 100 mels:

100 77 78 79 79 80 81 81 82 83 84

Here 100 mels = 77 cps, 101 mels = 78 cps, 105 mels = 81 cps, 109 mels = 84 cps, etc. As a second example, suppose one wished to find the frequency of a tone that is 100 mels higher than a 1600 cps tone. From the table we find that 1600 cps = 1343 mels. Then, 100 mels higher is 1443 mels which is equal to 1795 cps. Similarly the number of mel steps between two frequencies can be found.

It is quite feasible to transpose a passage in traditional musical notation to another register using equal mel intervals instead of equal frequency-ratio intervals. A frequency chart of the notes of the standard scale is required for this. Such charts may be found in many books on music theory, musical psychology, and musical acoustics. One may, for example, transpose the passage at Example 1a to begin on Great C, using intervals in the lower register which are equal in mels to those in the higher:

High Register

Note	Frequency	Mels	Mel Step Between Notes
c ³	1047 cps	1030	
db ³	1109 cps	1068	38 mels
f ³	1397 cps	1234	166 mels
g ³	1568 cps	1326	92 mels

Low Register

Mel Step	Mels	Frequency	Nearest Note of Scale
	83	65 cps	C
38 mels	121	92 cps	F [#]
166 mels	287	208 cps	g [#]
92 mels	379	277 cps	c ^{#1}

Thus the lower-register passage using intervals of the same mel size as the upper would be as in Example 1b. If the passage in Example 1c (the normal transposition) were played very rapidly the individual notes would not be perceptibly distinct, although the same passage played in the high register might be

very clear. The lower register passage with equal mel intervals (Example 1b), however, could be performed as rapidly as the higher and retain the same degree of clarity since the intervals of the lower passage are equally as perceptible as those of the higher. Further suggestions for the use of this scale will be mentioned below.

The present scale has some deficiencies which should be mentioned. First, the table produced by the computer analysis deviates somewhat at very low frequencies from the original curve. The region from 40 cps downward shows the greatest deviation but even this is slight when the data are plotted as mels vs. log frequency as in the original curve. The fact is that the data for the original curve given by Stevens and Volkman do not go below 40 cps. The portion of the curve below this was an extrapolation to a hypothetical zero pitch of 20 cps. A more accurate determination of the region below 40 cps would seem desirable although it may be questionable just how useful this region might be musically. In any event, the attention of the reader is drawn to this deficiency in the present table in the event that he might wish to use the table in this register.

Irrespective of the merits or defects of the present scale, it would seem profitable to have the mel scale determined by some other method than fractionation. It is well known that in psychological scaling, different methods which should logically give the same result do not always in fact do so. Tests are currently under way to determine whether a given size of mel interval, as given in the table, is judged the same size at the upper and lower extremes of the scale as it is in the middle. There is also a need to determine the form of the mel scale for different types of complex tones. (The present scale was constructed from tests involving sine tones only.) The author suspects that because of the presence of upper partials in complex tones, the mel scale for such tones would deviate significantly from the present scale, especially in the very low frequency region. This is not to say that the present table is not usable but only that refinements could be made if psychologists further investigated the matter.

That the mel scale could be put to use in musical composition seems evident. It opens the door to a new concept of interval, melody, and counterpoint. Melodies could be built of series of mel intervals, where equal intervals would be equal in subjective size rather than in frequency ratio. This leads to a counterpoint in which the various parts are equally perceptible in any register since they are constructed of equally perceptible

intervals. While it is evident that it is easiest to use the mel scale in electronic music where the frequency of tones is infinitely variable, it is also possible to apply the scale to traditional musical notation by using approximately equal mel steps as in Example 1. However, it is not the intention of this paper to attempt to discover all the uses of the mel scale but only to present the scale as a possible tool in musical composition.

Exactly what uses might be made of the mel scale in musical composition will depend entirely upon what use composers see fit to make of it. In an era in which composers are constantly looking for new modes of musical expression, this scale is worth some experimentation to see what possibilities it might present.

example

1



r e f e r e n c e s

- 1 S. S. Stevens, J. Volkman, and E. B. Newman, A Scale for the Measurement of the Psychological Magnitude of Pitch. In: *Journal of the Acoustical Society of America*, 8 (1937).
- 2 S. S. Stevens and J. Volkman, The Relation of Pitch to Frequency, A Revised Scale. In: *American Journal of Psychology*, 53 (1940).
- 3 R. Feldtekeller and E. Zwicker, *Das Ohr als Nachrichtenempfänger* (Stuttgart, 1956).
- 4 S. S. Stevens and H. Davis, *Hearing, Its Psychology and Physiology* (New York, 1938) 95.
- 5 Stevens and Davis, 94, 95, 97, 153.

table

1

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
0	2.	3.	4.	5.	5.	6.	7.	8.	9.	9.
10	10.	11.	12.	12.	13.	14.	15.	16.	16.	17.
20	18.	19.	20.	20.	21.	22.	23.	23.	24.	25.
30	26.	26.	27.	28.	29.	30.	30.	31.	32.	33.
40	33.	34.	35.	36.	36.	37.	38.	39.	39.	40.
50	41.	42.	42.	43.	44.	45.	45.	46.	47.	48.
60	48.	49.	50.	51.	51.	52.	53.	53.	54.	55.
70	56.	56.	57.	58.	59.	59.	60.	61.	61.	62.
80	63.	64.	64.	65.	66.	67.	67.	68.	69.	69.
90	70.	71.	72.	72.	73.	74.	74.	75.	76.	77.
100	77.	78.	79.	79.	80.	81.	81.	82.	83.	84.
110	84.	85.	86.	86.	87.	88.	89.	89.	90.	91.
120	91.	92.	93.	93.	94.	95.	96.	96.	97.	98.
130	98.	99.	100.	100.	101.	102.	103.	103.	104.	105.
140	105.	106.	107.	107.	108.	109.	109.	110.	111.	112.
150	112.	113.	114.	114.	115.	116.	116.	117.	118.	118.
160	119.	120.	120.	121.	122.	123.	123.	124.	125.	125.
170	126.	127.	127.	128.	129.	129.	130.	131.	132.	132.
180	133.	134.	134.	135.	136.	136.	137.	138.	138.	139.
190	140.	140.	141.	142.	143.	143.	144.	145.	145.	146.
200	147.	147.	148.	149.	149.	150.	151.	152.	152.	153.
210	154.	154.	155.	156.	156.	157.	158.	158.	159.	160.
220	160.	161.	162.	163.	163.	164.	165.	165.	166.	167.
230	167.	168.	169.	170.	170.	171.	172.	172.	173.	174.
240	174.	175.	176.	176.	177.	178.	179.	179.	180.	181.
250	181.	182.	183.	183.	184.	185.	186.	186.	187.	188.
260	188.	189.	190.	191.	191.	192.	193.	193.	194.	195.
270	195.	196.	197.	198.	198.	199.	200.	200.	201.	202.
280	203.	203.	204.	205.	205.	206.	207.	208.	208.	209.
290	210.	211.	211.	212.	213.	213.	214.	215.	216.	216.
300	217.	218.	218.	219.	220.	221.	221.	222.	223.	224.
310	224.	225.	226.	226.	227.	228.	229.	229.	230.	231.
320	232.	232.	233.	234.	235.	235.	236.	237.	238.	238.
330	239.	240.	241.	241.	242.	243.	244.	244.	245.	246.
340	247.	247.	248.	249.	250.	250.	251.	252.	253.	253.
350	254.	255.	256.	256.	257.	258.	259.	259.	260.	261.
360	262.	263.	263.	264.	265.	266.	266.	267.	268.	269.
370	269.	270.	271.	272.	273.	273.	274.	275.	276.	277.
380	277.	278.	279.	280.	280.	281.	282.	283.	284.	284.
390	285.	286.	287.	288.	288.	289.	290.	291.	292.	292.
400	293.	294.	295.	296.	296.	297.	298.	299.	300.	300.
410	301.	302.	303.	304.	305.	305.	306.	307.	308.	309.
420	309.	310.	311.	312.	313.	314.	314.	315.	316.	317.
430	318.	319.	319.	320.	321.	322.	323.	324.	324.	325.
440	326.	327.	328.	329.	330.	330.	331.	332.	333.	334.
450	335.	336.	336.	337.	338.	339.	340.	341.	342.	342.
460	343.	344.	345.	346.	347.	348.	348.	349.	350.	351.
470	352.	353.	354.	355.	356.	356.	357.	358.	359.	360.
480	361.	362.	363.	363.	364.	365.	366.	367.	368.	369.
490	370.	371.	372.	372.	373.	374.	375.	376.	377.	378.
500	379.	380.	381.	382.	383.	383.	384.	385.	386.	387.
510	388.	389.	390.	391.	392.	393.	394.	395.	395.	396.
520	397.	398.	399.	400.	401.	402.	403.	404.	405.	406.
530	407.	408.	409.	410.	411.	412.	412.	413.	414.	415.

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
540	416.	417.	418.	419.	420.	421.	422.	423.	424.	425.
550	426.	427.	428.	429.	430.	431.	432.	433.	434.	435.
560	436.	437.	438.	439.	440.	441.	442.	443.	444.	445.
570	446.	447.	448.	449.	450.	451.	452.	453.	454.	455.
580	456.	457.	458.	459.	460.	461.	462.	463.	464.	465.
590	466.	467.	468.	469.	470.	471.	472.	473.	474.	475.
600	476.	477.	478.	479.	480.	482.	483.	484.	485.	486.
610	487.	488.	489.	490.	491.	492.	493.	494.	495.	496.
620	497.	498.	500.	501.	502.	503.	504.	505.	506.	507.
630	508.	509.	510.	511.	512.	514.	515.	516.	517.	518.
640	519.	520.	521.	522.	523.	524.	526.	527.	528.	529.
650	530.	531.	532.	533.	534.	536.	537.	538.	539.	540.
660	541.	542.	543.	544.	546.	547.	548.	549.	550.	551.
670	552.	553.	555.	556.	557.	558.	559.	560.	561.	563.
680	564.	565.	566.	567.	568.	570.	571.	572.	573.	574.
690	575.	576.	578.	579.	580.	581.	582.	583.	585.	586.
700	587.	588.	589.	591.	592.	593.	594.	595.	596.	598.
710	599.	600.	601.	602.	604.	605.	606.	607.	608.	610.
720	611.	612.	613.	614.	616.	617.	618.	619.	620.	622.
730	623.	624.	625.	627.	628.	629.	630.	631.	633.	634.
740	635.	636.	638.	639.	640.	641.	642.	644.	645.	646.
750	647.	649.	650.	651.	652.	654.	655.	656.	657.	659.
760	660.	661.	662.	664.	665.	666.	668.	669.	670.	671.
770	673.	674.	675.	676.	678.	679.	680.	682.	683.	684.
780	685.	687.	688.	689.	691.	692.	693.	694.	696.	697.
790	698.	700.	701.	702.	704.	705.	706.	707.	709.	710.
800	711.	713.	714.	715.	717.	718.	719.	721.	722.	723.
810	725.	726.	727.	729.	730.	731.	733.	734.	735.	737.
820	738.	739.	741.	742.	743.	745.	746.	747.	749.	750.
830	751.	753.	754.	755.	757.	758.	759.	761.	762.	764.
840	765.	766.	768.	769.	770.	772.	773.	775.	776.	777.
850	779.	780.	781.	783.	784.	786.	787.	788.	790.	791.
860	793.	794.	795.	797.	798.	799.	801.	802.	804.	805.
870	806.	808.	809.	811.	812.	814.	815.	816.	818.	819.
880	821.	822.	823.	825.	826.	828.	829.	831.	832.	833.
890	835.	836.	838.	839.	841.	842.	843.	845.	846.	848.
900	849.	851.	852.	854.	855.	856.	858.	859.	861.	862.
910	864.	865.	867.	868.	870.	871.	873.	874.	875.	877.
920	878.	880.	881.	883.	884.	886.	887.	889.	890.	892.
930	893.	895.	896.	898.	899.	901.	902.	904.	905.	907.
940	908.	910.	911.	913.	914.	916.	917.	919.	920.	922.
950	923.	925.	926.	928.	929.	931.	932.	934.	935.	937.
960	938.	940.	941.	943.	944.	946.	947.	949.	950.	952.
970	953.	955.	957.	958.	960.	961.	963.	964.	966.	967.
980	969.	970.	972.	974.	975.	977.	978.	980.	981.	983.
990	984.	986.	987.	989.	991.	992.	994.	995.	997.	998.
1000	1000.	1002.	1003.	1005.	1006.	1008.	1009.	1011.	1013.	1014.
1010	1016.	1017.	1019.	1020.	1022.	1024.	1025.	1027.	1028.	1030.
1020	1032.	1033.	1035.	1036.	1038.	1040.	1041.	1043.	1044.	1046.
1030	1048.	1049.	1051.	1052.	1054.	1056.	1057.	1059.	1060.	1062.
1040	1064.	1065.	1067.	1069.	1070.	1072.	1073.	1075.	1077.	1078.
1050	1080.	1082.	1083.	1085.	1086.	1088.	1090.	1091.	1093.	1095.
1060	1096.	1098.	1099.	1101.	1103.	1104.	1106.	1108.	1109.	1111.
1070	1113.	1114.	1116.	1118.	1119.	1121.	1123.	1124.	1126.	1128.

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1080	1129.	1131.	1133.	1134.	1136.	1138.	1139.	1141.	1143.	1144.
1090	1146.	1148.	1149.	1151.	1153.	1154.	1156.	1158.	1159.	1161.
1100	1163.	1164.	1166.	1168.	1169.	1171.	1173.	1174.	1176.	1178.
1110	1179.	1181.	1183.	1185.	1186.	1188.	1190.	1191.	1193.	1195.
1120	1196.	1198.	1200.	1202.	1203.	1205.	1207.	1208.	1210.	1212.
1130	1214.	1215.	1217.	1219.	1220.	1222.	1224.	1226.	1227.	1229.
1140	1231.	1232.	1234.	1236.	1238.	1239.	1241.	1243.	1245.	1246.
1150	1248.	1250.	1251.	1253.	1255.	1257.	1258.	1260.	1262.	1264.
1160	1265.	1267.	1269.	1271.	1272.	1274.	1276.	1278.	1279.	1281.
1170	1283.	1285.	1286.	1288.	1290.	1292.	1293.	1295.	1297.	1299.
1180	1300.	1302.	1304.	1306.	1308.	1309.	1311.	1313.	1315.	1316.
1190	1318.	1320.	1322.	1323.	1325.	1327.	1329.	1331.	1332.	1334.
1200	1336.	1338.	1339.	1341.	1343.	1345.	1347.	1348.	1350.	1352.
1210	1354.	1356.	1357.	1359.	1361.	1363.	1365.	1366.	1368.	1370.
1220	1372.	1374.	1375.	1377.	1379.	1381.	1383.	1384.	1386.	1388.
1230	1390.	1392.	1393.	1395.	1397.	1399.	1401.	1403.	1404.	1406.
1240	1408.	1410.	1412.	1413.	1415.	1417.	1419.	1421.	1423.	1424.
1250	1426.	1428.	1430.	1432.	1434.	1435.	1437.	1439.	1441.	1443.
1260	1445.	1446.	1448.	1450.	1452.	1454.	1456.	1457.	1459.	1461.
1270	1463.	1465.	1467.	1469.	1470.	1472.	1474.	1476.	1478.	1480.
1280	1482.	1483.	1485.	1487.	1489.	1491.	1493.	1495.	1496.	1498.
1290	1500.	1502.	1504.	1506.	1508.	1509.	1511.	1513.	1515.	1517.
1300	1519.	1521.	1523.	1524.	1526.	1528.	1530.	1532.	1534.	1536.
1310	1538.	1539.	1541.	1543.	1545.	1547.	1549.	1551.	1553.	1555.
1320	1556.	1558.	1560.	1562.	1564.	1566.	1568.	1570.	1572.	1574.
1330	1575.	1577.	1579.	1581.	1583.	1585.	1587.	1589.	1591.	1593.
1340	1594.	1596.	1598.	1600.	1602.	1604.	1606.	1608.	1610.	1612.
1350	1614.	1615.	1617.	1619.	1621.	1623.	1625.	1627.	1629.	1631.
1360	1633.	1635.	1637.	1639.	1640.	1642.	1644.	1646.	1648.	1650.
1370	1652.	1654.	1656.	1658.	1660.	1662.	1664.	1666.	1668.	1669.
1380	1671.	1673.	1675.	1677.	1679.	1681.	1683.	1685.	1687.	1689.
1390	1691.	1693.	1695.	1697.	1699.	1701.	1703.	1705.	1707.	1708.
1400	1710.	1712.	1714.	1716.	1718.	1720.	1722.	1724.	1726.	1728.
1410	1730.	1732.	1734.	1736.	1738.	1740.	1742.	1744.	1746.	1748.
1420	1750.	1752.	1754.	1756.	1758.	1760.	1762.	1764.	1766.	1768.
1430	1769.	1771.	1773.	1775.	1777.	1779.	1781.	1783.	1785.	1787.
1440	1789.	1791.	1793.	1795.	1797.	1799.	1801.	1803.	1805.	1807.
1450	1809.	1811.	1813.	1815.	1817.	1819.	1821.	1823.	1825.	1827.
1460	1829.	1831.	1833.	1835.	1837.	1839.	1841.	1843.	1845.	1847.
1470	1849.	1851.	1853.	1855.	1857.	1859.	1861.	1864.	1866.	1868.
1480	1870.	1872.	1874.	1876.	1878.	1880.	1882.	1884.	1886.	1888.
1490	1890.	1892.	1894.	1896.	1898.	1900.	1902.	1904.	1906.	1908.
1500	1910.	1912.	1914.	1916.	1918.	1920.	1922.	1924.	1926.	1929.
1510	1931.	1933.	1935.	1937.	1939.	1941.	1943.	1945.	1947.	1949.
1520	1951.	1953.	1955.	1957.	1959.	1961.	1963.	1965.	1968.	1970.
1530	1972.	1974.	1976.	1978.	1980.	1982.	1984.	1986.	1988.	1990.
1540	1992.	1994.	1996.	1999.	2001.	2003.	2005.	2007.	2009.	2011.
1550	2013.	2015.	2017.	2019.	2021.	2023.	2026.	2028.	2030.	2032.
1560	2034.	2036.	2038.	2040.	2042.	2044.	2046.	2049.	2051.	2053.
1570	2055.	2057.	2059.	2061.	2063.	2065.	2067.	2070.	2072.	2074.
1580	2076.	2078.	2080.	2082.	2084.	2086.	2088.	2091.	2093.	2095.
1590	2097.	2099.	2101.	2103.	2105.	2107.	2110.	2112.	2114.	2116.
1600	2118.	2120.	2122.	2124.	2127.	2129.	2131.	2133.	2135.	2137.
1610	2139.	2141.	2144.	2146.	2148.	2150.	2152.	2154.	2156.	2159.

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1620	2161.	2163.	2165.	2167.	2169.	2171.	2174.	2176.	2178.	2180.
1630	2182.	2184.	2186.	2189.	2191.	2193.	2195.	2197.	2199.	2202.
1640	2204.	2206.	2208.	2210.	2212.	2214.	2217.	2219.	2221.	2223.
1650	2225.	2227.	2230.	2232.	2234.	2236.	2238.	2240.	2243.	2245.
1660	2247.	2249.	2251.	2254.	2256.	2258.	2260.	2262.	2264.	2267.
1670	2269.	2271.	2273.	2275.	2278.	2280.	2282.	2284.	2286.	2289.
1680	2291.	2293.	2295.	2297.	2300.	2302.	2304.	2306.	2308.	2311.
1690	2313.	2315.	2317.	2319.	2322.	2324.	2326.	2328.	2330.	2333.
1700	2335.	2337.	2339.	2342.	2344.	2346.	2348.	2350.	2353.	2355.
1710	2357.	2359.	2362.	2364.	2366.	2368.	2371.	2373.	2375.	2377.
1720	2379.	2382.	2384.	2386.	2388.	2391.	2393.	2395.	2397.	2400.
1730	2402.	2404.	2406.	2409.	2411.	2413.	2415.	2418.	2420.	2422.
1740	2424.	2427.	2429.	2431.	2434.	2436.	2438.	2440.	2443.	2445.
1750	2447.	2449.	2452.	2454.	2456.	2459.	2461.	2463.	2465.	2468.
1760	2470.	2472.	2475.	2477.	2479.	2481.	2484.	2486.	2488.	2491.
1770	2493.	2495.	2497.	2500.	2502.	2504.	2507.	2509.	2511.	2514.
1780	2516.	2518.	2521.	2523.	2525.	2527.	2530.	2532.	2534.	2537.
1790	2539.	2541.	2544.	2546.	2548.	2551.	2553.	2555.	2558.	2560.
1800	2562.	2565.	2567.	2569.	2572.	2574.	2576.	2579.	2581.	2583.
1810	2586.	2588.	2590.	2593.	2595.	2598.	2600.	2602.	2605.	2607.
1820	2609.	2612.	2614.	2616.	2619.	2621.	2624.	2626.	2628.	2631.
1830	2633.	2635.	2638.	2640.	2643.	2645.	2647.	2650.	2652.	2654.
1840	2657.	2659.	2662.	2664.	2666.	2669.	2671.	2674.	2676.	2678.
1850	2681.	2683.	2686.	2688.	2690.	2693.	2695.	2698.	2700.	2703.
1860	2705.	2707.	2710.	2712.	2715.	2717.	2720.	2722.	2724.	2727.
1870	2729.	2732.	2734.	2737.	2739.	2741.	2744.	2746.	2749.	2751.
1880	2754.	2756.	2759.	2761.	2764.	2766.	2768.	2771.	2773.	2776.
1890	2778.	2781.	2783.	2786.	2788.	2791.	2793.	2796.	2798.	2801.
1900	2803.	2806.	2808.	2811.	2813.	2816.	2818.	2821.	2823.	2826.
1910	2828.	2831.	2833.	2836.	2838.	2841.	2843.	2846.	2848.	2851.
1920	2853.	2856.	2858.	2861.	2863.	2866.	2868.	2871.	2873.	2876.
1930	2879.	2881.	2884.	2886.	2889.	2891.	2894.	2896.	2899.	2901.
1940	2904.	2907.	2909.	2912.	2914.	2917.	2919.	2922.	2925.	2927.
1950	2930.	2932.	2935.	2937.	2940.	2943.	2945.	2948.	2950.	2953.
1960	2956.	2958.	2961.	2963.	2966.	2969.	2971.	2974.	2976.	2979.
1970	2982.	2984.	2987.	2990.	2992.	2995.	2997.	3000.	3003.	3005.
1980	3008.	3011.	3013.	3016.	3019.	3021.	3024.	3027.	3029.	3032.
1990	3035.	3037.	3040.	3043.	3045.	3048.	3051.	3053.	3056.	3059.
2000	3061.	3064.	3067.	3069.	3072.	3075.	3077.	3080.	3083.	3086.
2010	3088.	3091.	3094.	3096.	3099.	3102.	3105.	3107.	3110.	3113.
2020	3115.	3118.	3121.	3124.	3126.	3129.	3132.	3135.	3137.	3140.
2030	3143.	3146.	3148.	3151.	3154.	3157.	3159.	3162.	3165.	3168.
2040	3171.	3173.	3176.	3179.	3182.	3184.	3187.	3190.	3193.	3196.
2050	3198.	3201.	3204.	3207.	3210.	3213.	3215.	3218.	3221.	3224.
2060	3227.	3230.	3232.	3235.	3238.	3241.	3244.	3247.	3249.	3252.
2070	3255.	3258.	3261.	3264.	3267.	3269.	3272.	3275.	3278.	3281.
2080	3284.	3287.	3290.	3293.	3295.	3298.	3301.	3304.	3307.	3310.
2090	3313.	3316.	3319.	3322.	3325.	3328.	3330.	3333.	3336.	3339.
2100	3342.	3345.	3348.	3351.	3354.	3357.	3360.	3363.	3366.	3369.
2110	3372.	3375.	3378.	3381.	3384.	3387.	3390.	3393.	3396.	3399.
2120	3402.	3405.	3408.	3411.	3414.	3417.	3420.	3423.	3426.	3429.
2130	3432.	3435.	3438.	3441.	3444.	3447.	3450.	3453.	3456.	3459.
2140	3462.	3466.	3469.	3472.	3475.	3478.	3481.	3484.	3487.	3490.
2150	3493.	3496.	3500.	3503.	3506.	3509.	3512.	3515.	3518.	3521.

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
2160	3525.	3528.	3531.	3534.	3537.	3540.	3543.	3547.	3550.	3553.
2170	3556.	3559.	3562.	3566.	3569.	3572.	3575.	3578.	3582.	3585.
2180	3588.	3591.	3594.	3598.	3601.	3604.	3607.	3611.	3614.	3617.
2190	3620.	3624.	3627.	3630.	3633.	3637.	3640.	3643.	3646.	3650.
2200	3653.	3656.	3660.	3663.	3666.	3669.	3673.	3676.	3679.	3683.
2210	3686.	3689.	3693.	3696.	3699.	3703.	3706.	3709.	3713.	3716.
2220	3719.	3723.	3726.	3730.	3733.	3736.	3740.	3743.	3747.	3750.
2230	3753.	3757.	3760.	3764.	3767.	3770.	3774.	3777.	3781.	3784.
2240	3788.	3791.	3794.	3798.	3801.	3805.	3808.	3812.	3815.	3819.
2250	3822.	3826.	3829.	3833.	3836.	3840.	3843.	3847.	3850.	3854.
2260	3857.	3861.	3864.	3868.	3872.	3875.	3879.	3882.	3886.	3889.
2270	3893.	3897.	3900.	3904.	3907.	3911.	3915.	3918.	3922.	3925.
2280	3929.	3933.	3936.	3940.	3944.	3947.	3951.	3954.	3958.	3962.
2290	3965.	3969.	3973.	3977.	3980.	3984.	3988.	3991.	3995.	3999.
2300	4002.	4006.	4010.	4014.	4017.	4021.	4025.	4029.	4032.	4036.
2310	4040.	4044.	4048.	4051.	4055.	4059.	4063.	4066.	4070.	4074.
2320	4078.	4082.	4086.	4089.	4093.	4097.	4101.	4105.	4109.	4113.
2330	4116.	4120.	4124.	4128.	4132.	4136.	4140.	4144.	4148.	4152.
2340	4155.	4159.	4163.	4167.	4171.	4175.	4179.	4183.	4187.	4191.
2350	4195.	4199.	4203.	4207.	4211.	4215.	4219.	4223.	4227.	4231.
2360	4235.	4239.	4243.	4247.	4251.	4255.	4260.	4264.	4268.	4272.
2370	4276.	4280.	4284.	4288.	4292.	4296.	4301.	4305.	4309.	4313.
2380	4317.	4321.	4325.	4330.	4334.	4338.	4342.	4346.	4351.	4355.
2390	4359.	4363.	4367.	4372.	4376.	4380.	4384.	4389.	4393.	4397.
2400	4401.	4406.	4410.	4414.	4419.	4423.	4427.	4432.	4436.	4440.
2410	4445.	4449.	4453.	4458.	4462.	4466.	4471.	4475.	4480.	4484.
2420	4488.	4493.	4497.	4502.	4506.	4510.	4515.	4519.	4524.	4528.
2430	4533.	4537.	4542.	4546.	4551.	4555.	4560.	4564.	4569.	4573.
2440	4578.	4582.	4587.	4591.	4596.	4601.	4605.	4610.	4614.	4619.
2450	4624.	4628.	4633.	4637.	4642.	4647.	4651.	4656.	4661.	4665.
2460	4670.	4675.	4679.	4684.	4689.	4693.	4698.	4703.	4708.	4712.
2470	4717.	4722.	4727.	4731.	4736.	4741.	4746.	4751.	4755.	4760.
2480	4765.	4770.	4775.	4779.	4784.	4789.	4794.	4799.	4804.	4809.
2490	4814.	4819.	4823.	4828.	4833.	4838.	4843.	4848.	4853.	4858.
2500	4863.	4868.	4873.	4878.	4883.	4888.	4893.	4898.	4903.	4908.
2510	4913.	4918.	4923.	4928.	4933.	4939.	4944.	4949.	4954.	4959.
2520	4964.	4969.	4974.	4980.	4985.	4990.	4995.	5000.	5005.	5011.
2530	5016.	5021.	5026.	5032.	5037.	5042.	5047.	5053.	5058.	5063.
2540	5068.	5074.	5079.	5084.	5090.	5095.	5100.	5106.	5111.	5116.
2550	5122.	5127.	5133.	5138.	5143.	5149.	5154.	5160.	5165.	5171.
2560	5176.	5182.	5187.	5193.	5198.	5204.	5209.	5215.	5220.	5226.
2570	5231.	5237.	5242.	5248.	5254.	5259.	5265.	5270.	5276.	5282.
2580	5287.	5293.	5299.	5304.	5310.	5316.	5321.	5327.	5333.	5339.
2590	5344.	5350.	5356.	5362.	5367.	5373.	5379.	5385.	5391.	5396.
2600	5402.	5408.	5414.	5420.	5426.	5432.	5437.	5443.	5449.	5455.
2610	5461.	5467.	5473.	5479.	5485.	5491.	5497.	5503.	5509.	5515.
2620	5521.	5527.	5533.	5539.	5545.	5551.	5557.	5563.	5569.	5576.
2630	5582.	5588.	5594.	5600.	5606.	5613.	5619.	5625.	5631.	5637.
2640	5644.	5650.	5656.	5662.	5669.	5675.	5681.	5687.	5694.	5700.
2650	5706.	5713.	5719.	5726.	5732.	5738.	5745.	5751.	5757.	5764.
2660	5770.	5777.	5783.	5790.	5796.	5803.	5809.	5816.	5822.	5829.
2670	5835.	5842.	5848.	5855.	5862.	5868.	5875.	5881.	5888.	5895.
2680	5901.	5908.	5915.	5921.	5928.	5935.	5942.	5948.	5955.	5962.
2690	5969.	5975.	5982.	5989.	5996.	6003.	6009.	6016.	6023.	6030.
2700	6037.	6044.	6051.	6058.	6065.	6072.	6079.	6085.	6092.	6099.
2710	6106.	6113.	6120.	6128.	6135.	6142.	6149.	6156.	6163.	6170.

MELS	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
2720	6177.	6184.	6191.	6199.	6206.	6213.	6220.	6227.	6235.	6242.
2730	6249.	6256.	6264.	6271.	6278.	6285.	6293.	6300.	6307.	6315.
2740	6322.	6329.	6337.	6344.	6352.	6359.	6367.	6374.	6381.	6389.
2750	6396.	6404.	6411.	6419.	6427.	6434.	6442.	6449.	6457.	6464.
2760	6472.	6480.	6487.	6495.	6503.	6510.	6518.	6526.	6533.	6541.
2770	6549.	6557.	6564.	6572.	6580.	6588.	6596.	6604.	6611.	6619.
2780	6627.	6635.	6643.	6651.	6659.	6667.	6675.	6683.	6691.	6699.
2790	6707.	6715.	6723.	6731.	6739.	6747.	6755.	6763.	6771.	6780.
2800	6788.	6796.	6804.	6812.	6820.	6829.	6837.	6845.	6853.	6862.
2810	6870.	6878.	6887.	6895.	6903.	6912.	6920.	6928.	6937.	6945.
2820	6954.	6962.	6971.	6979.	6988.	6996.	7005.	7013.	7022.	7036.
2830	7039.	7047.	7056.	7065.	7073.	7082.	7091.	7099.	7108.	7117.
2840	7126.	7134.	7143.	7152.	7161.	7169.	7178.	7187.	7196.	7205.
2850	7214.	7223.	7231.	7240.	7249.	7258.	7267.	7276.	7285.	7294.
2860	7303.	7312.	7321.	7330.	7340.	7349.	7358.	7367.	7376.	7385.
2870	7394.	7404.	7413.	7422.	7431.	7441.	7450.	7459.	7469.	7478.
2880	7487.	7497.	7506.	7515.	7525.	7534.	7544.	7553.	7563.	7572.
2890	7582.	7591.	7601.	7610.	7620.	7629.	7639.	7649.	7658.	7668.
2900	7678.	7687.	7697.	7707.	7716.	7726.	7736.	7746.	7756.	7765.
2910	7775.	7785.	7795.	7805.	7815.	7825.	7835.	7845.	7855.	7865.
2920	7875.	7885.	7895.	7905.	7915.	7925.	7935.	7945.	7955.	7965.
2930	7976.	7986.	7996.	8006.	8016.	8027.	8037.	8047.	8058.	8068.
2940	8078.	8089.	8099.	8109.	8120.	8130.	8141.	8151.	8162.	8172.
2950	8183.	8193.	8204.	8215.	8225.	8236.	8246.	8257.	8268.	8278.
2960	8289.	8300.	8311.	8321.	8332.	8343.	8354.	8365.	8376.	8386.
2970	8397.	8408.	8419.	8430.	8441.	8452.	8463.	8474.	8485.	8496.
2980	8507.	8518.	8530.	8541.	8552.	8563.	8574.	8585.	8597.	8608.
2990	8619.	8631.	8642.	8653.	8665.	8676.	8687.	8699.	8710.	8722.
3000	8733.	8745.	8756.	8768.	8779.	8791.	8802.	8814.	8825.	8837.
3010	8849.	8860.	8872.	8884.	8896.	8907.	8919.	8931.	8943.	8955.
3020	8967.	8978.	8990.	9002.	9014.	9026.	9038.	9050.	9062.	9074.
3030	9086.	9098.	9110.	9123.	9135.	9147.	9159.	9171.	9184.	9196.
3040	9208.	9220.	9233.	9245.	9257.	9270.	9282.	9295.	9307.	9319.
3050	9332.	9344.	9357.	9369.	9382.	9395.	9407.	9420.	9433.	9445.
3060	9458.	9471.	9483.	9496.	9509.	9522.	9534.	9547.	9560.	9573.
3070	9586.	9599.	9612.	9625.	9638.	9651.	9664.	9677.	9690.	9703.
3080	9716.	9729.	9743.	9756.	9769.	9782.	9795.	9809.	9822.	9835.
3090	9849.	9862.	9876.	9889.	9902.	9916.	9929.	9943.	9956.	9970.
3100	9983.	9997.	10011.	10024.	10038.	10052.	10065.	10079.	10093.	10107.